Koblitz cover transfers — R28

Adjugate norm functions select actual ordinary Hecke images

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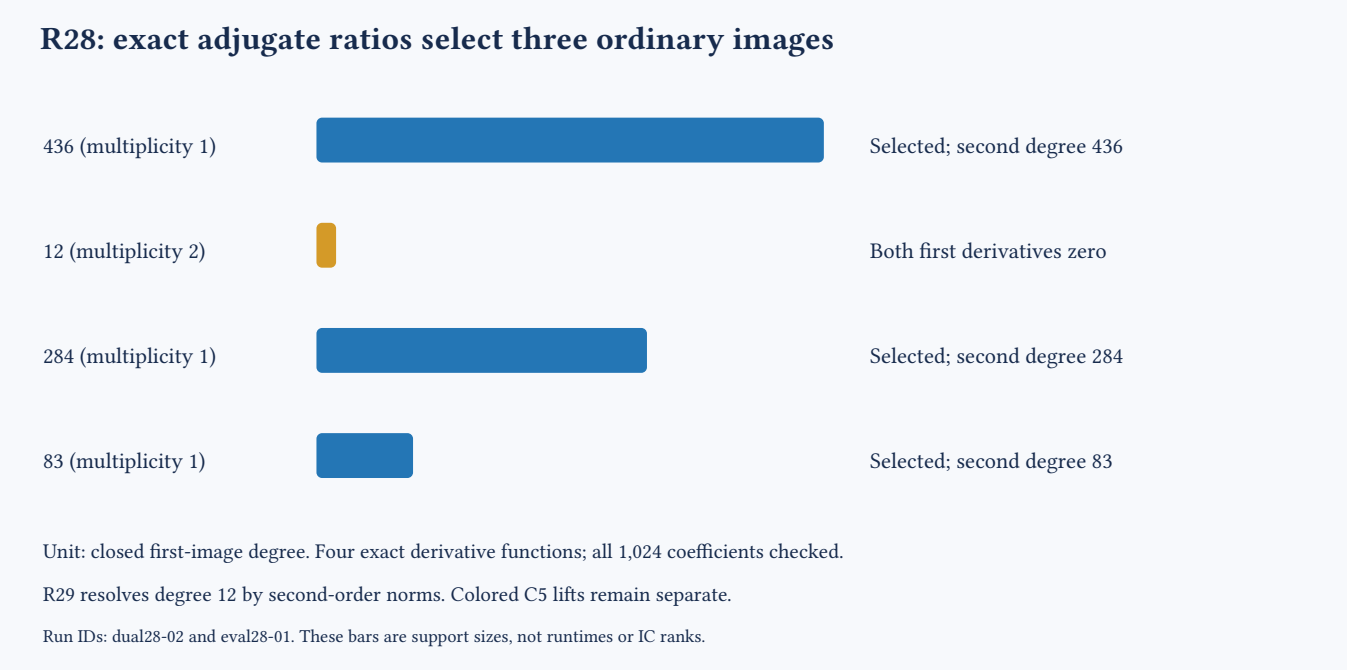
The elliptic equation alone leaves three rational second-x candidates at each of four ordinary supports. R28 uses the actual 132-branch Hecke norm to select images, under the retained modularity, good-reduction, pole and RR bridge.

For epsilon²=0, write N(F+epsilon H)=N(F)+epsilon D_H. At a simple zero with D_1 nonzero, the ratio D_x/D_1 gives the actual second x-coordinate. The finite-factor algebra is proved in Lean. Its identification with the actual modular norm and fiber remains written. The norm as multiplication determinant is reviewed in Sutherland, Definition 7.1.



Native Rust extends the original cyclic-121 binary-series determinant to dual numbers, including ten middle branches and the large branch. Every dual inverse and planted multiplicativity check passes. All underlying norms reproduce the original M15 A/B coefficients exactly.

The four derivative functions are recovered in the original degree-792/660 base RR spaces. All 1,024 coefficients of every function are checked, using 123,904 input coefficients. Python performs only assembly and metadata.

Selected images and precise remaining scope

Function	First degree	Multiplicity	R28 outcome
y0·yinf+1	436	1	Selected, degree 436
y0·yinf+1	12	2	Both first adjugates zero
x0·yinf+1	284	1	Selected, degree 284
x0·yinf+1	83	1	Selected, degree 83

Every selected point satisfies the actual target cubic and matches exactly one of its three rational roots. Full Frobenius orbit checks establish the second-image degrees and residue pushforward multiplicity one. The zero at degree 12 is kept as a failure of first-order recovery. R29 resolves it with second-order norms; it is not divided away here.

The raw archive contains the exact A/B derivative functions and field-coordinate point pairs. These are paired Hecke images. They are not complete scalar-C5 source fibers, exceptional fibers, arbitrary reverse divisor evaluations, actual modular-factor actions or independent relation rows.

The first compiler attempt fails before mathematical execution because a copied inner doc comment is placed after a definition. Its source and log are retained. The corrected build and both native runs pass.

Lean compiles three named theorems and audits four declarations, including one generated congruence theorem. There are zero failures and zero trusted native results. The proofs formalize cofactor nonvanishing and weighted finite-factor recovery; no actual modular inclusion is assumed as an axiom.

R29′s complete dependency check subsequently identifies the degree-436 and degree-12 product-plus-one pairs as half-orbit Frobenius conjugates. Their base closed-divisor difference cancels, while the colored C5 deck offset remains a separate construction. No IC rank or efficiency conclusion is promoted from these image calculations.